

Moving-container verification report

Executed on 4 October 2026. Synthetic numerical verification of declared rigid mechanics; no experimental material validation or universal best-solver claim.

The archive retains **53 histories**. Each setting has one warm-up and 3 recorded timing samples. All 54 Python/research tests pass. Native comparators are pinned Box2D 2.4.1 (block) and 3.1.1 (temporal). This study does not execute Vektor, WASM or physical GPU. Timing measures solver plus controller, excluding Python, JSON, state readout and process startup. Results are from one Linux machine, not a controlled cross-platform throughput study.

Independent exact packed-row check

N touching unit-mass disks initially rest inside a closed box translating at 1 m/s. For the declared rigid, zero-restitution, frictionless constraints, every outgoing ball must move at 1 m/s. The first-output-frame maximum individual error budget is 0.01 m/s. Expected momentum is N kg m/s, kinetic energy N/2 J, actuator work N J and dissipation N/2 J. Native first-frame failure includes contact discovery and floating geometry; it cannot be attributed solely to iterations. A physical elastic chain would transmit finite-speed waves.

Balls	Setting	Max velocity error (m/s)	Budget met	Median solver+controller (ms)
4	block_p1_s1	1	no	0.02028
4	block_p8_s32	0.0025841	yes	0.06809
4	block_p32_s128	1.431e-06	yes	0.6326
4	temporal_p8_s32	1.2517e-05	yes	0.258
16	block_p1_s1	1	no	0.03167
16	block_p8_s32	0.590995	no	0.2273
16	block_p32_s128	0.103714	no	3.006
16	temporal_p8_s32	0.209123	no	0.4029
64	block_p1_s1	1	no	0.07278
64	block_p8_s32	1	no	0.5702
64	block_p32_s128	1	no	9.514
64	temporal_p8_s32	0.996492	no	1.361
100	block_p1_s1	1	no	0.107
100	block_p8_s32	1	no	1.049
100	block_p32_s128	1	no	14.82
100	temporal_p8_s32	1	no	2.467

The independent dense frozen normal projection satisfies all four rows with maximum velocity error 5.68e-14 m/s and checked complementarity residuals. This is a correctness oracle, not a complete frictional engine or a measured performance winner.

Frictional 100-ball motion

Each scene lasts 2 s, with gravity -9.81 m/s², radius 0.1 m, mass 1 kg, friction 0.4 and restitution 0. Four wall fixtures move as one kinematic body. Motion is translation at 0.6 m/s, reversals every 0.5 s, or rotation at 0.5 rad/s. These coefficients are idealized declarations. Fine simulation is not physical ground truth.

Reference qualification requires every adjacent edge in two separate sweeps to meet quarter budgets: position RMS 0.005 m, velocity RMS 0.0125 m/s and spin RMS 0.0125 rad/s. Primary updates are 16/32/64 at 128 iterations; iterations are 32/64/128 at 64 primary updates. The highest-work state is only a candidate reference unless all four edges pass.

Motion	Reference qualified	Worst refinement / quarter budget
translate	no	45.2

Motion	Reference qualified	Worst refinement / quarter budget
shake	no	96.7
rotate	no	40.72

Motion	Setting	Position RMS (m)	Velocity RMS (m/s)	Spin RMS (rad/s)	Qualified accuracy pass
translate	block_p1_s1	0.124	0.3281	1.398	unqualified
translate	block_p1_s8	0.06287	0.1727	1.071	unqualified
translate	block_p4_s16	0.03521	0.094	0.7361	unqualified
translate	block_p8_s32	0.02784	0.07779	0.7241	unqualified
translate	temporal_p4_s16	0.0865	0.1257	1.065	unqualified
translate	block_adaptive	0.03617	0.1106	0.937	unqualified
shake	block_p1_s1	0.1243	0.4826	2.276	unqualified
shake	block_p1_s8	0.05912	0.2965	1.863	unqualified
shake	block_p4_s16	0.02256	0.2055	1.397	unqualified
shake	block_p8_s32	0.02378	0.1687	1.344	unqualified
shake	temporal_p4_s16	0.07096	0.1882	1.251	unqualified
shake	block_adaptive	0.02842	0.2049	1.402	unqualified
rotate	block_p1_s1	0.1337	0.2902	1.701	unqualified
rotate	block_p1_s8	0.05092	0.1516	0.9723	unqualified
rotate	block_p4_s16	0.0152	0.08127	0.6194	unqualified
rotate	block_p8_s32	0.0158	0.07317	0.5944	unqualified
rotate	temporal_p4_s16	0.09754	0.1351	1.021	unqualified
rotate	block_adaptive	0.02168	0.08344	0.6575	unqualified

Candidate budgets are 0.02 m, 0.05 m/s and 0.05 rad/s. Comparisons against an unqualified reference are diagnostic and cannot establish accuracy. No tolerance was relaxed after looking at the results. The current adaptive controller is a whole-world heuristic; this study does not certify error-controlled adaptation.

Controls and geometric diagnostics

Control, 100 balls	Max position difference (m)	Max velocity difference	
		(m/s)	Max spin difference (rad/s)
galilean_100	0.001254	0.002916	0
comoving_100	3.624e-06	2.38e-08	0
reverse_body_order_100	0.0009751	0.006165	0

Controls cover a Galilean boost, joint free translation of box and contents, and reversing body insertion order. Differences are reported rather than hidden. Containment is checked in the actual rotating box frame. Independent disk-pair distances and wall extents are checked at every output frame.

Motion	High setting max disk overlap (m)	High setting max wall violation (m)
translate	0.001032	0.0006252
shake	0.001581	0.001498
rotate	0.00102	0.0007355

Observed geometry does not prove continuous no-tunneling. Contents momentum is not conserved under moving walls. The constant-translation, zero-gravity row and control runs include actuator-work accounting. Shaking and rotation need a future time-resolved contact reaction ledger; no exact work result is claimed for them.

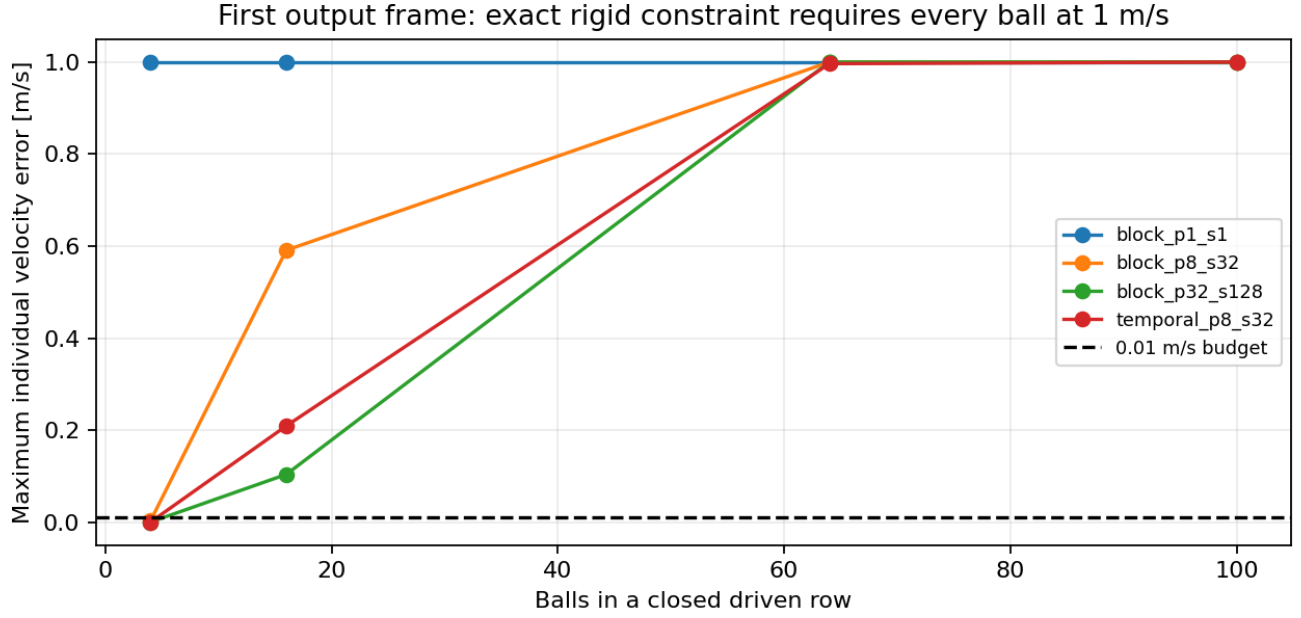


Figure 1: Packed-row error

Consequences for the engine

The packed-row failures falsify a universal accuracy claim for the current high preset. Spend work within one unchanged physical law, preserving contact/history states, and investigate sparse globally coupled island solves. The normal oracle establishes the target constraints but does not yet supply a complete Coulomb friction solver. Combine convergence, independent analytic tests, contact/CCD geometry and energy plus actuator work; residuals alone do not bound future trajectory error.

An accuracy order through discontinuous impacts has not been established. Use per-observable tolerances and event timing, and test smooth-region integration order separately. Authentic elastic tangential/rolling rebound needs measured response and stored contact history. Coarse deformable cells require additional degrees of freedom, not merely material labels on a rigid body. The established contact algebra and coefficient-based friction model remain nonnovel; the joint publication verdict remains conditional.

Reproduction and evidence

Run `python -m research.audit_container_study` to independently check the archive, all trajectory comparisons, analytical row metrics and reference gates. The plan, source, summary, failures and every full trace are retained. See `README.md` for build/reproduction commands and `contact-model.pdf` for typeset mathematics.

Execution source: 6a805dbbf940b2ba448c006d031db4f201a4e1ff. Archive SHA-256: a96c4e19d500b043abdb52b8571b65cc22f

Sources supporting the method choices: Catto, Solver2D, Box2D simulation, MuJoCo contact computation. Prior-art references and opposing reviews are retained in `research/critical-review.md`, `publication-case.md` and `joint-verdict.md`. No reference is used to turn synthetic coefficients into experimental measurements.